

MUHAMMAD ASAD LAL

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ABOUT ME

I am currently pursuing a Master’s degree in Computer Science and Technology at Nanjing University of Science and Technology. I have experience in programming, research, and technical problem-solving, with a strong willingness to learn and adapt in a professional environment. I aim to contribute to cutting-edge technologies and have a positive impact on both organizational and societal settings.

ACADEMIC QUALIFICATION

- Master of Science in Computer Science & Technology

Nanjing University of Science and Technology, China

Research: “Optimization using Metaheuristic Algorithms”

Supervisor: Prof. Bo Huang

Sep 2024 - Apr 2027
- Bachelor of Science in Computer Science, CGPA 3.82/4.00 (Top in Class)

Government College University Faisalabad, Pakistan

Final Year Project: “Heart Disease Prediction (Android Application)”

Supervisor: Prof. Ali Raza Shah

Aug 2019 - Sep 2023

PUBLICATIONS

1. Muhammad Asad Lal, Muhammad Kashif, Bo Huang*. “Father-inspired metaheuristic: a hybrid framework with neighborhood repulsion and elite learning for continuous optimization.”

Evolutionary Intelligence, Vol. 19, Article 111, 2026.

DOI: 10.1007/s12065-026-01228-w
2. Muhammad Asad Lal, Muhammad Kashif, JianYong Lv, Bo Huang*. “Forman-Ricci-Curvature-Guided SHADE for Multi-Level Thresholding in Fundus Image Segmentation.”

Applied Soft Computing, 2026. Submitted.

RESEARCH WORK

- Optimization using Metaheuristic Algorithms

Nanjing University of Science and Technology, China

Sep 2024 - Present

- Designed and implemented new metaheuristic algorithms in MATLAB and Python.
 - Proposed the Father-Inspired Metaheuristic algorithm for continuous optimization.
 - Developed Forman-Ricci-Curvature-Guided SHADE for multi-level thresholding in fundus image segmentation
 - Evaluated algorithm performance using benchmarks, convergence analysis, and quality metrics.

RESEARCH DIRECTIONS

- Optimization Algorithms
 - Image Processing
 - Pattern Recognition
- Engineering Design Optimization
 - Machine Learning/Artificial Intelligence
 - Deep Learning Models

PROFESSIONAL EXPERIENCE

- Android Development, Internship

Idealogic Software House

Oct 2023 - Mar 2024

- Contributed to Android mobile app development projects.
 - Worked with the team during the app development lifecycle.
 - Improved practical skills in Android development, programming, and software practices.

UNDERGRADUATE PROJECTS

1. **Heart Disease Prediction, Final Year Project** Aug 2019 - Sep 2023
Developed an AI-based Android app using Logistic Regression to classify users' heart health as "Healthy" or "Unhealthy" and provide basic recommendations.
2. **Deep Learning-based Face Recognition, Semester Project** Feb 2023 - May 2023
Developed a CNN-based face recognition model trained on image data to identify faces and patterns.

PROFESSIONAL AND PERSONAL SKILLS

- **Programming Languages:** Python, MATLAB, Java, and C++.
- **Optimization & Research:** Metaheuristic algorithms, Continuous optimization, and Statistical analysis.
- **Machine Learning & AI:** Machine learning models, data analysis, and AI-based problem solving.
- **Image Processing:** Medical image segmentation, multi-level thresholding, and image quality evaluation.
- **Android Development:** Android app development, API integration, Java, and Android Studio.
- **Problem-solving & Adaptability:** Strong analytical thinking, Quick learning, and Research-oriented.
- **Communication & Writing:** Academic writing, technical documentation, and international article writing.
www.israelhayom.com/writer/muhammad-asad-lal

AWARDS & CERTIFICATIONS

One Million Prompters, Certification <i>Dubai Future Foundation</i>	June 24, 2026
Chinese Writing Competition, Award Certificate <i>Nanjing University of Science and Technology</i>	Dec 31, 2024
Manage Sensitive Data and Security Policies within GitHub, Certification <i>Microsoft</i>	Apr 01, 2024
Outstanding Academic Performance Award <i>Government College University Faisalabad</i>	Sep 29, 2023
Machine Learning with Python, Certification <i>International Business Machines (IBM)</i>	Nov 22, 2023
Deep Learning with TensorFlow, Certification <i>International Business Machines (IBM)</i>	Nov 06, 2023
Computer Application and Office Professional, Certification <i>Government of Punjab Vocational Training Council</i>	Oct 17, 2017

EXTRA-CURRICULAR ACTIVITIES

Basic Community Safety Training <i>Punjab Emergency Service Rescue 1122, Pakistan</i>	Jun 15, 2023
BSAFE Security Awareness Training <i>United Nations Department of Safety and Security</i>	May 26, 2022

REFERENCE

Will be furnished on Request.